

BALA KRISHNA MAKINENI

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EDUCATION

B.Tech in Computer Science and Engineering, RVR & JC College of Engineering
CGPA: 8.1 / 10.0

2020 – 2024

TECHNICAL SKILLS

Languages: Java, Python, SQL

Frameworks: Spring Boot, Hibernate, FastApi, React.js

Databases: PostgreSQL, Redis

Cloud/Tools: AWS (S3, EC2, SQS), Git, GitHub Actions

Core CS: MicroServices, DSA, OOPs, DBMS

EXPERIENCE

Software Development Engineer in Test - I

Jul 2024 – Dec 2025

Bidgely, Bengaluru, India

- Engineered asynchronous data preparation modules using Java multithreading for mock and prod API datasets.
- Automated P0/P1 test cases within the core framework, enhancing critical regression coverage and reducing QA validation time, by detecting critical functional bugs.
- Contributed to SQS-based asynchronous test orchestration, optimising distributed execution and raising test pass rate to 90%.
- Integrated CI/CD automation via GitHub Actions to auto-trigger validation suites for each deployment cycle.
- Contributed to backend automation frameworks using Java, Spring Boot, AWS SQS, and TestNG, enabling scalable and reducing the test suite execution time to 35%.

Backend Developer Intern

Feb 2024 – Jun 2024

NAATS Aviation Pvt Ltd, Guntur, India

- Built REST APIs in Node.js and Express.js to handle telemetry data at 50ms intervals, deployed on AWS EC2.
- Collaborated with cross-functional teams, including Aeronautical Domain experts, to refine project requirements.
- Implemented Redis Pub/Sub for faster real-time data communication and reduced inter-service latency.

PROJECTS

PDF RAG Chatbot ([Code](#))

Python, FastAPI, Langchain, VectorDB, LLM, Embedding

- Built a PDF RAG API using FastAPI, LangChain, Groq LLM (LLaMA 3.3-70B), and ChromaDB for semantic document Q&A.
- Designed modular layered architecture separating API routes, services, RAG pipeline, and schemas for scalability.
- Implemented HuggingFace sentence-transformer embeddings with persistent ChromaDB for multi-document ingestion.
- Developed an interactive Jinja2 web interface with PDF upload, chat-based querying, and source attribution display.

Finance Tracker ([BE](#) — [FE](#))

Java, Spring-Boot, Spring Data-Jpa, Spring Security, PostgreSQL, React.js

- Developed a Spring Boot-based Finance Tracker backend to manage user transactions with debit/credit support.
- Implemented JWT authentication using Spring Security for secure user registration, login, and API access.
- Built RESTful APIs with Spring Data JPA to handle transaction persistence and time-based aggregations(daily/monthly).
- Developed an interactive React.js dashboard with line and bar charts to visualize monthly and yearly financial insights.

ACHIEVEMENTS & CERTIFICATIONS

- Awarded **Spot Award** at Bidgely for Automating the P0/P1 cases at framework level to detect functional bugs.
- Solved 350+ DSA problems on LeetCode and GeeksforGeeks
- Winner, **InformaTrix Coding Contest** at RVR & JC College of Engineering.
- Certified in **Web Development (Udemy)**, **Programming in Java (NPTEL)**, **Python (MTA Microsoft)**, **AWS Cloud Foundations**.